

# Syed Maoz Arif

+92 321 914 3144 | syedmaoza@gmail.com | linkedin.com/in/syed-maoz | buismaoz346-bit.github.io/my-website | Abbottabad, KPK, Pakistan

## PROFESSIONAL SUMMARY

Electrical Engineering undergraduate specialising in PCB design, embedded systems, and IoT. Altium Global Scholarship recipient (2026) and Instructables Sensors Contest Runner-Up (2026). Experienced freelance hardware design engineer delivering DFM-ready PCBs and 3D assemblies to international clients. Eager to apply proven hardware expertise in a professional engineering role.

## EDUCATION

- B.Sc. Electrical (Electronics) Engineering — COMSATS University Islamabad, Abbottabad**

Expected 2028
- Coursework: Digital Logic Design, Embedded Systems, Circuit Theory, Signals & Systems.
  - Altium Global Scholarship for Emerging Engineers (2026)** — awarded internationally to outstanding hardware design students.

## TECHNICAL SKILLS

|                |                                                                                             |
|----------------|---------------------------------------------------------------------------------------------|
| PCB Design     | Altium Designer, Altium 365 — multi-layer routing, schematic capture, DRC, Gerber output    |
| Embedded / IoT | ESP32, Arduino Nano, STM32, MPU6050; C/C++ firmware; Firebase real-time data streaming      |
| CAD / 3D       | Autodesk Fusion 360, Inventor — parametric modelling, assembly design, 3D-print preparation |
| Programming    | C/C++, Python (OOP, automation, simulation), LaTeX / Overleaf                               |

## PROJECTS & RECOGNITION

- Omni-Sense Node | ESP32 · Altium Designer · Firebase · C++**

2025
- Centralised DAQ & Control System on a single ESP32 integrating 15 peripherals with real-time Firebase telemetry.
  - Designed and fabricated a multi-layer PCB; implemented sensor polling, data fusion, and remote monitoring firmware.
- IncliSense | ESP32 · MPU6050 · Altium Designer · C++ — Runner-Up, Instructables Sensors Contest 2026**

2026
- High-precision digital inclinometer with sub-degree resolution using MPU6050 IMU and real-time tilt-compensation.
  - Delivered custom PCB, 3D-printed enclosure, and full documentation; recognised internationally on Instructables.

## EXPERIENCE & LEADERSHIP

- Freelance PCB Design Engineer — Fiverr & Upwork**

2024 – Present
- End-to-end PCB design for international clients: schematic capture, multi-layer routing, DFM review, Gerber delivery.
  - Parametric 3D product models and manufacturing-ready assemblies using Autodesk Fusion 360 and Inventor.
- Membership Head — IET On Campus, Abbottabad**

2024 – Present
- Directed student recruitment and technical seminars, expanding active membership and industry engagement.
- Cabinet Member — Youth Parliament Pakistan**

2025 – 2027
- Represent KPK in national policy simulations; drafted resolutions on technology, education, and youth development.